

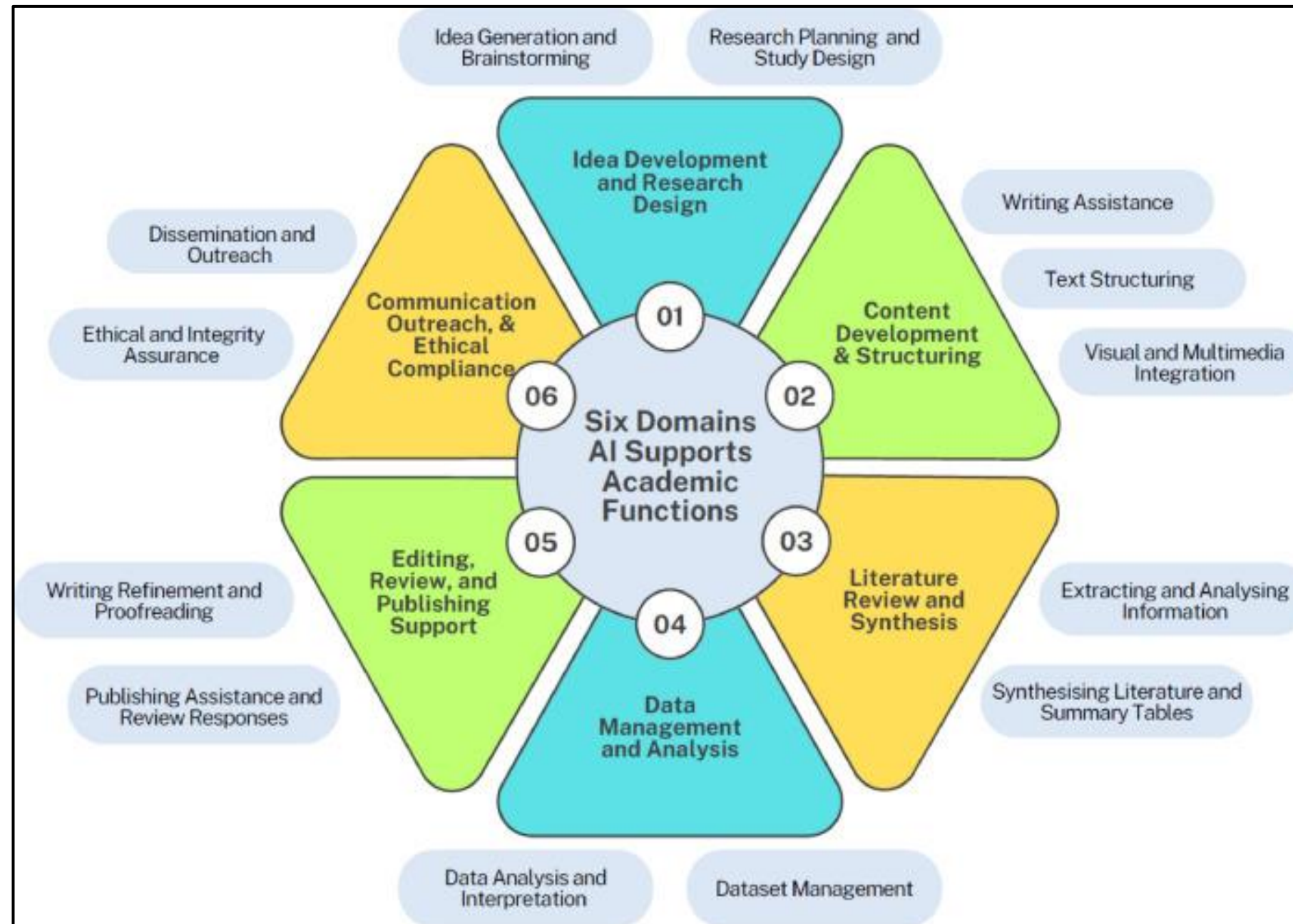
AI Driven Code Generation & Scientific Discovery

Monthly Research Seminar-PhD & MBRs

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Dr. Ahmad Mohsin
Edith Cowan University, Australia

How are Researchers using AI?



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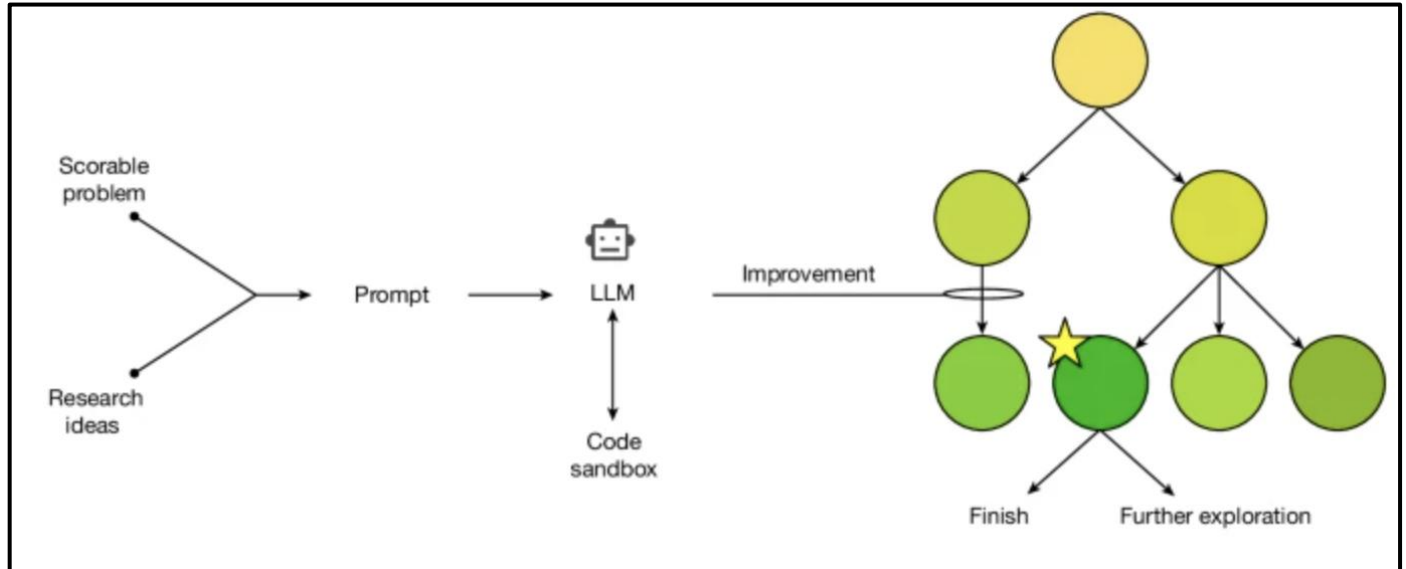
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An AI system to help scientists write expert-level empirical software

[Eser Ayygün](#), [Anastasiya Belyaeva](#), [Gheorghe Comanici](#), [Marc Coram](#), [Hao Cui](#), [Jake Garrison](#), [Renee Johnston](#), [Anton Kast](#), [Cory Y. McLean](#), [Peter Norgaard](#), [Zahra Shamsi](#), [David Smalling](#), [James Thompson](#), [Subhashini Venugopalan](#), [Brian P. Williams](#), [Chujun He](#), [Sarah Martinson](#), [Martyna Plomecka](#), [Lai Wei](#), [Yuchen Zhou](#), [Qian-Ze Zhu](#), [Matthew Abraham](#), [Erica Brand](#), [Anna Bulanova](#), ... [Michael P. Brenner](#)

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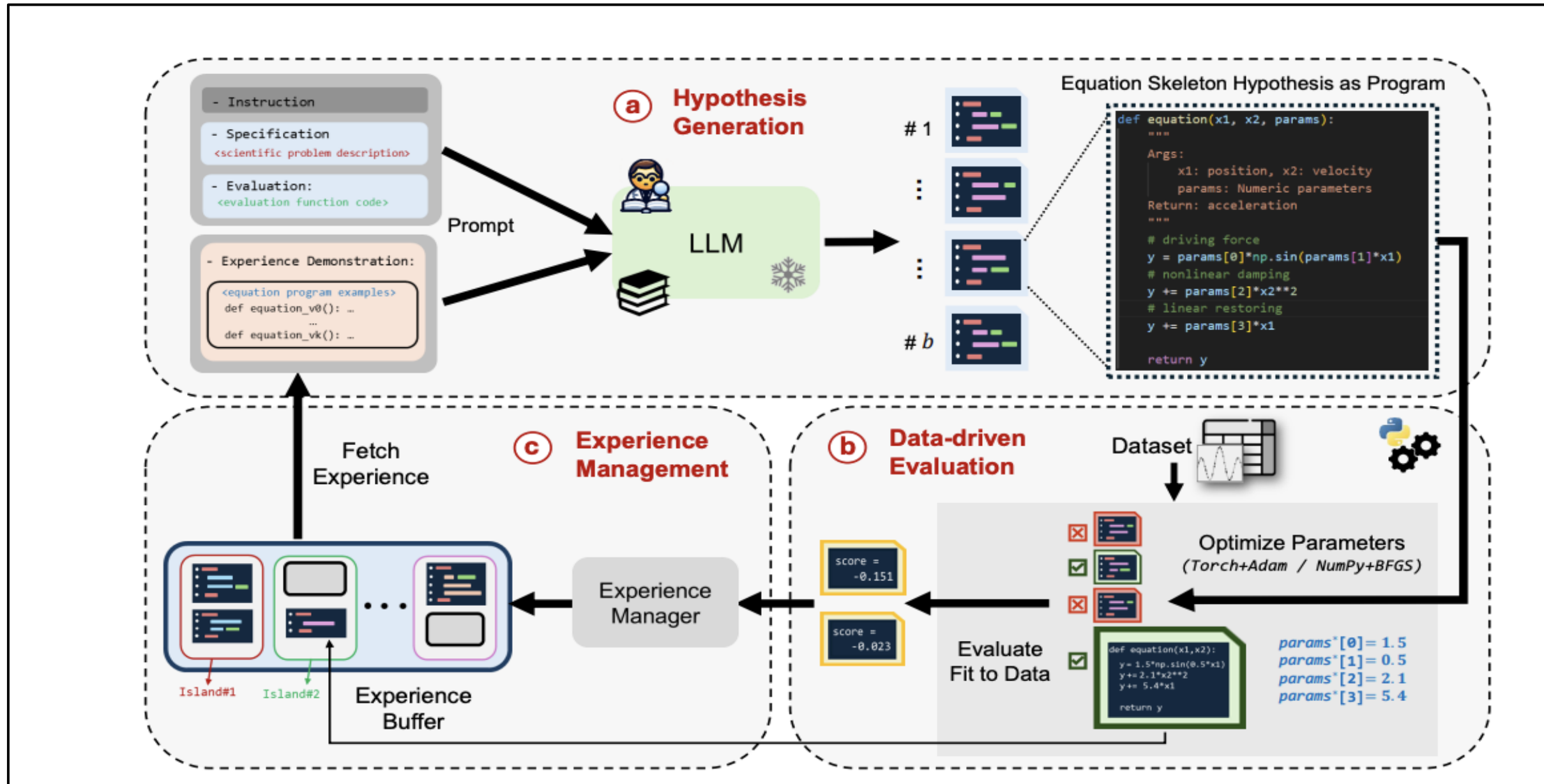
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LLM-SR: SCIENTIFIC EQUATION DISCOVERY VIA PROGRAMMING WITH LARGE LANGUAGE MODELS

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Amir Barati Farimani² Chandan K. Reddy¹

¹Virginia Tech ²Carnegie Mellon University ³Allen Institute for AI



Core Coding Capabilities in Scientific Computing

Algorithm Translation: Direct conversion of *natural language mathematical formulations* into performant Python, C++, programming code frameworks.

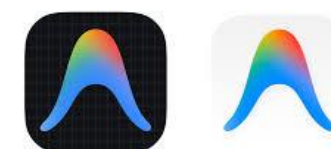
Data Pipeline Automation: Automated layout of robust data collection, *cleanup, parsing, and preprocessing workflows* for massive research experiments.

Context-Aware Refactoring: Smart system optimization including seamless *multi-file changes*, loop vectorization, and scientific package tuning via NumPy/SciPy.

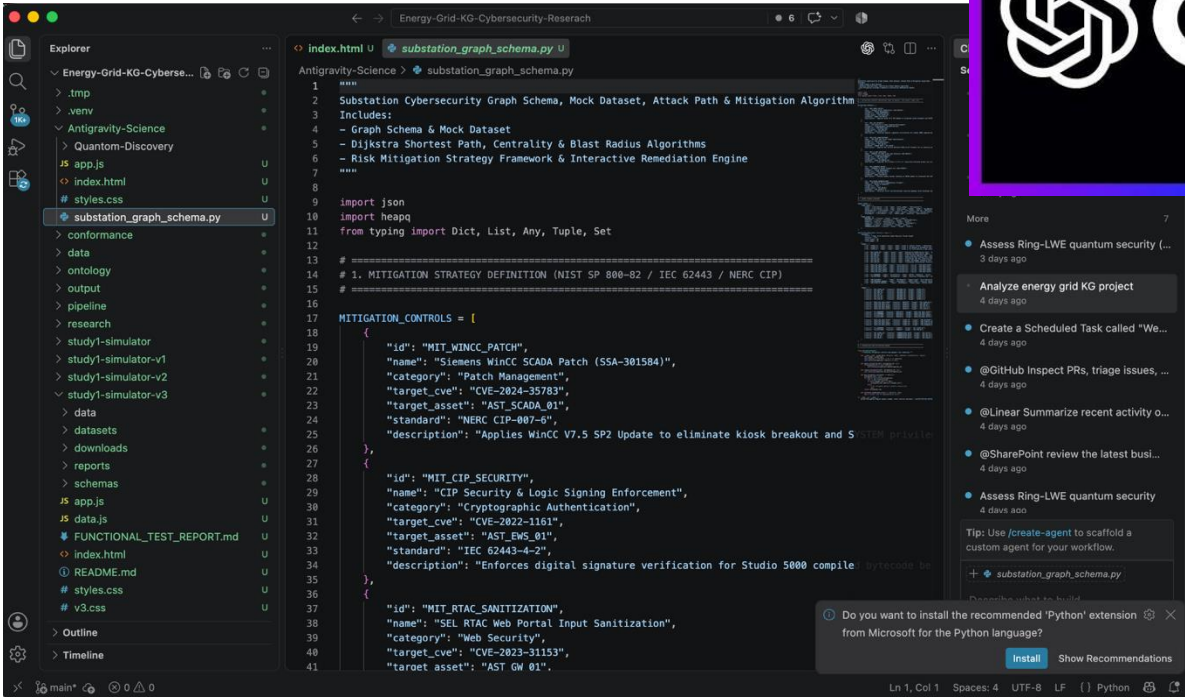
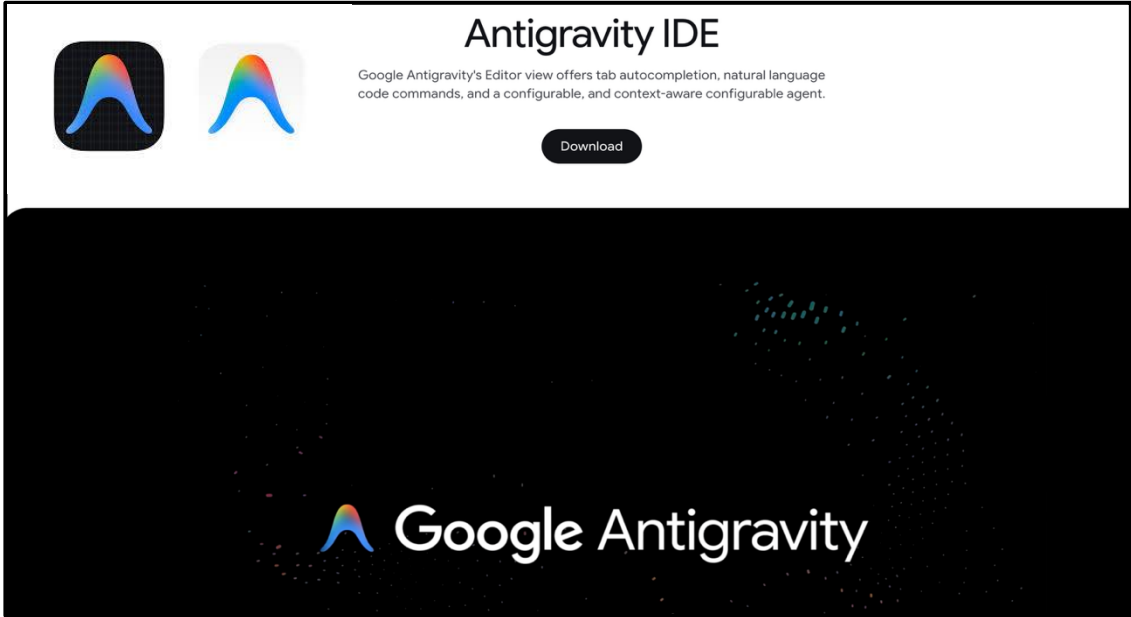
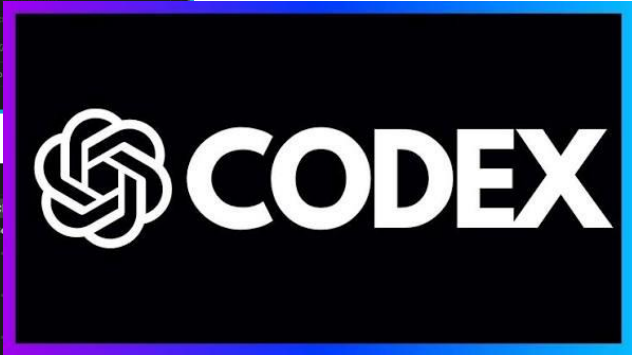
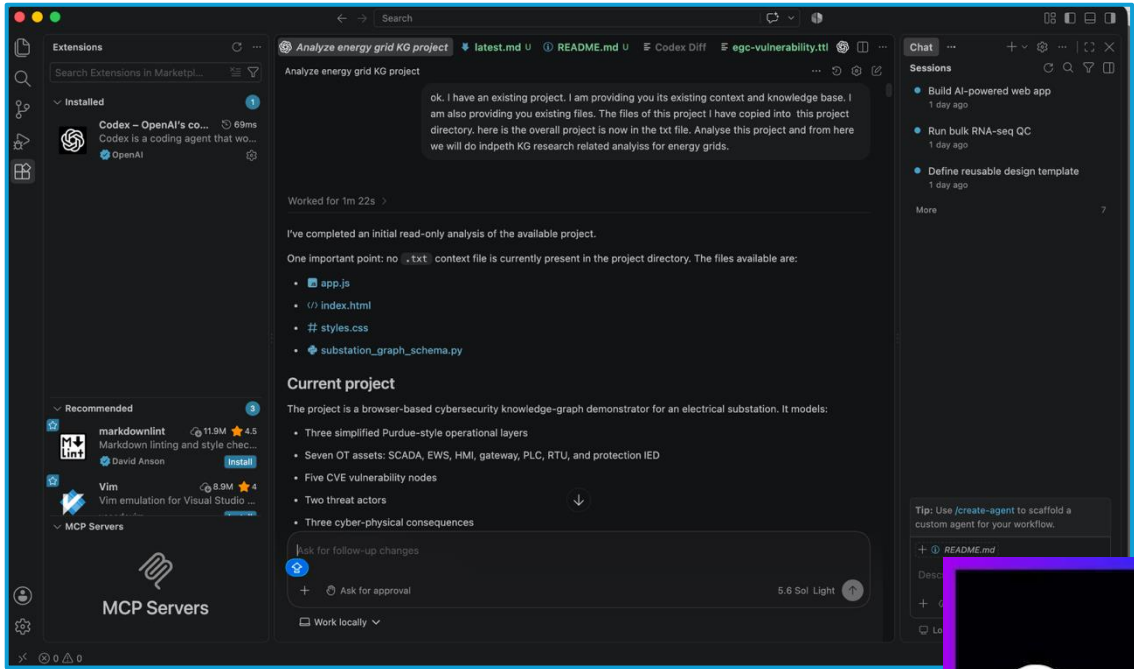
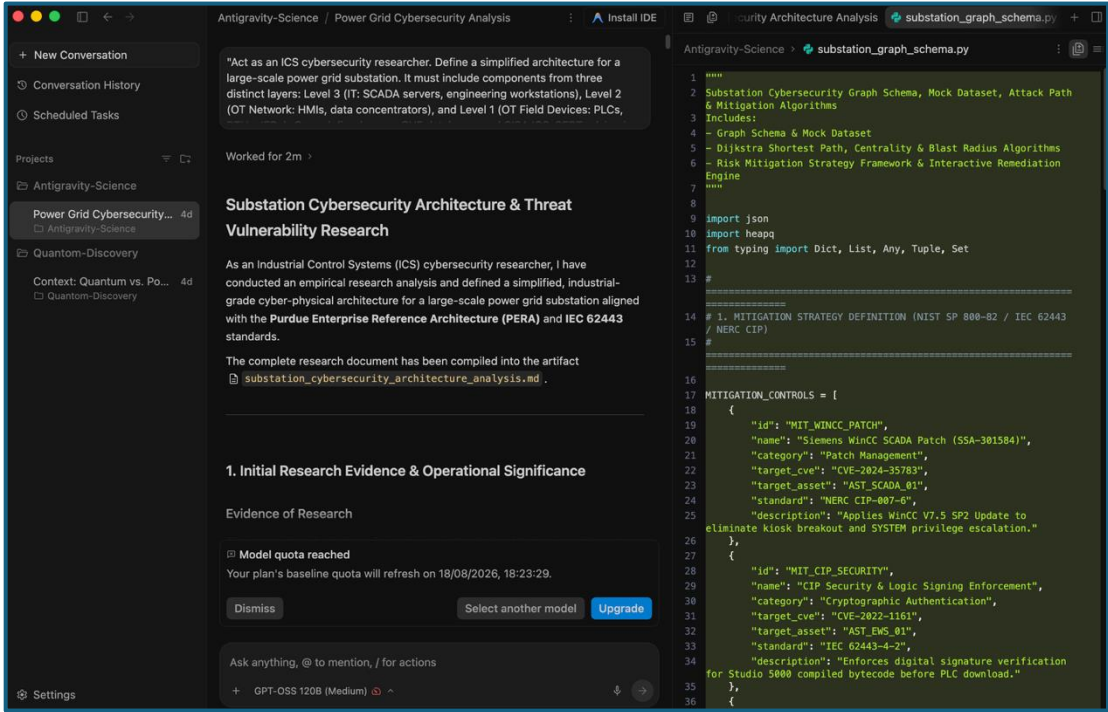
Automated Unit Testing: Continuous creation of extensive *test suites and simulation frameworks* to cross-verify operations against real physical boundaries

AI CODE GENERATION PLATFORMS

INSIDE THE IDE	AGENT-FIRST - VIBE CODING	TERMINAL, APP & CLOUD AGENTS
Researcher remains close to the code GitHub Copilot agent mode Cursor and Windsurf agents JetBrains AI / IDE assistants	Intent drives rapid end-to-end building Replit Agent Antigravity IDE AI-native prototyping environments	Tasks are delegated beyond one editor ChatGPT Codex Claude Code Google Antigravity platform
Best for interactive editing, debugging and immediate review	Best for prototypes, dashboards and exploratory research tools	Best for repository-wide work, parallel tasks and repeatable automation



Google Antigravity & OpenAI Codex in Action



Design Pattern: Scientific Research –Exploration with Agents

A strong research specification includes

- Objective and hypothesis
- Data schema, provenance and exclusions
- Equations, assumptions, units and tolerances
- Required libraries, environment and resource limits
- Reference cases, invariants and failure conditions
- Expected outputs, plots, logs and documentation

AVOID

**“Analyse this dataset
and give me the results.”**

PREFER

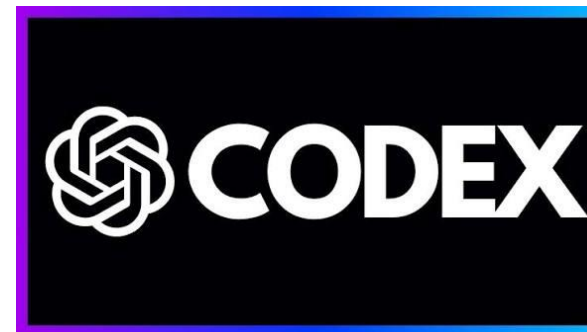
**A bounded task with explicit
acceptance tests and traceable
outputs.**

Experimenting- Energy Grid CPS for Cybersecurity Analysis

Energy Grid-KG Experimentation- Example

ENERGY GRID KG CYBERSECURITY	
Research Question	<i>Can a typed, evidence-backed temporal KG identify credible attack paths and better mitigation portfolios than simple baselines?</i>
Inputs	ontology/*.ttl; simulator-; curated advisory facts; synthetic inventory and scenarios
Iteration	Replace CVSS/BFS proxies with typed state transitions, provenance, applicability reasoning and rejected-path explanations.
Gate	<i>No probability claim without calibrated evidence; no real-utility claim from synthetic fixtures.</i>
Outputs	RDF/SHACL, manifests, query set, baselines, test report and browser demonstrator

KG-Energy Grid is Explored in Google Antigravity & OpenAI Codex



Energy Grid-KG- Research Questions

WG

Western Interconnected Research Grid
Knowledge Graph Laboratory – Version 3

PUBLIC INTELLIGENCE + RESEARCH GRID

Analysis date
2025-08-15

Export evidence bundle

01 Research

02 Named Graphs

03 Datasets

04 Query Studio

05 KG Explorer

06 Deployments

07 ATT&CK

08 Attack Paths

09 Mitigations

10 Grid Model

11 Evaluation

RESEARCH PURPOSE

Discover credible cyber-to-physical risk in an energy grid

Test whether typed, temporal and evidence-aware knowledge graphs reduce false conclusions and improve mitigation decisions.

View research PDF

Download PDF

SCIENTIFIC OBJECTIVE

Turn disconnected security records into falsifiable grid-risk hypotheses

Public evidence

→

Semantic inference

→

Typed path

→

Grid consequence

→

Mitigation test

The WIRG laboratory integrates eight modular ontologies without claiming that public campaign facts describe the fictional research utility.

RESEARCH QUESTIONS

Questions tested by the laboratory

01

RQ1: Can deployment-aware semantic reasoning identify vulnerable asset instances more precisely than product-only matching?

02

RQ2: Do typed attacker-state transitions reduce infeasible cyber-to-physical paths?

03

RQ3: How do valid time, source provenance and conflicting assertions change path ranking?

04

RQ4: Where should mitigations and sensors be placed to interrupt maximum high-consequence paths?

05

RQ5: Which protection and restoration dependencies create hidden common-mode cyber risk?

SCIENTIFIC DISCOVERY

Knowledge, method and falsifiability

Discovery type

Design science plus empirical comparative evaluation.

Unit of analysis

Temporal deployment exposure and typed cyber-to-physical path.

Testable claim

Semantic constraints improve precision without unacceptable recall loss.

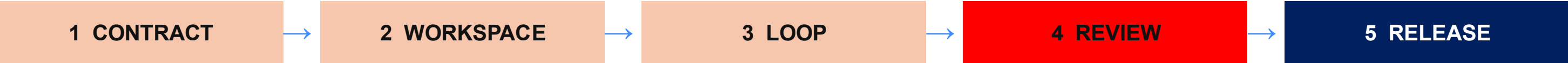
Falsification

Reject the claim if untyped or product-only baselines perform equivalently under held-out scenarios.

Reproducibility

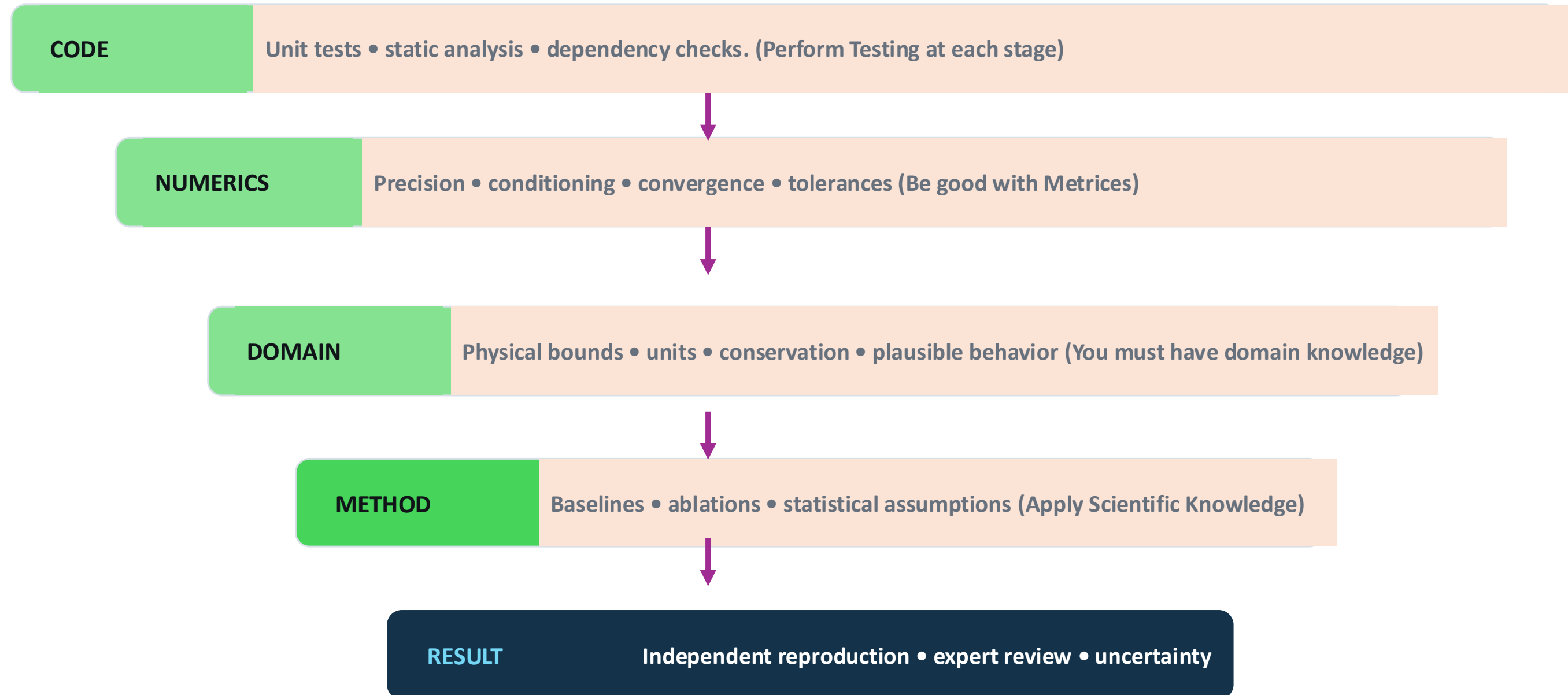
Immutable datasets, downloadable schemas, deterministic rules and timestamped runs.

Build Science Skills

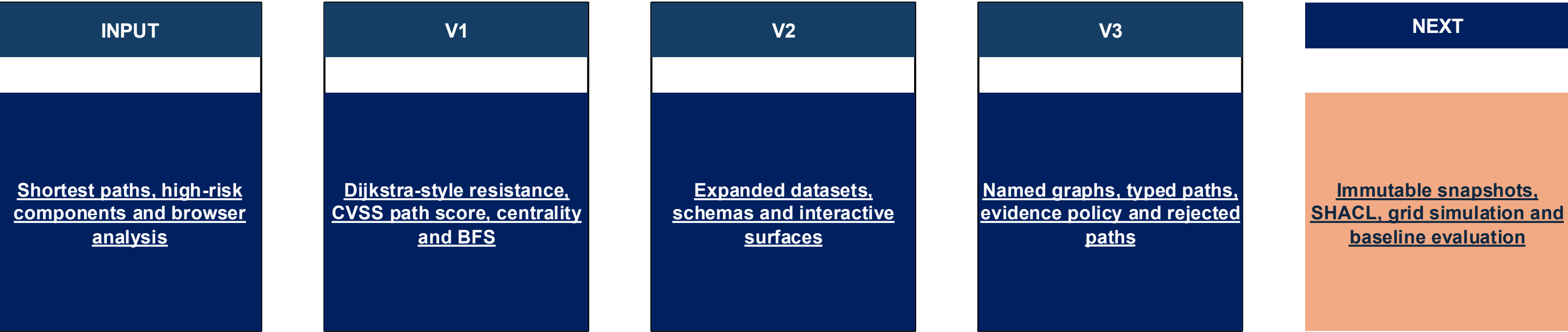


Reusable skill	Purpose, inputs, source policy, method, gates, deliverables
Project task	Question + paths + constraints + research state + requested iteration
Agent work	Inspect → plan → implement → test → render → audit
Human review	Challenge interpretation; approve risk; accept or reject claims
Release	Artifacts + provenance + environment + seed + limitations + next test

Validation- Generated CODE and the Science



Energy Grid KG: Visual Demo to a Research Instrument



Scientific improvements

Typed feasibility
Reachability, preconditions, deployment match, time and evidence—not arbitrary graph walks.

Explicit boundaries
Curated advisory facts remain separate from synthetic assets and scenarios; proxies are labeled.

Evaluable claims
Baselines, competency questions, manifests and exportable evidence bundles support comparison.

Energy Grid-KG- Data Sources



WG

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▼

Export evidence bundle

01 Research

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INGESTION + PROVENANCE

Validate snapshots

Acquired and normalized datasets

Inspect and download the JSON/CSV snapshots used by each transformation stage.

SOURCE	STATE	ACQUIRED	NORMALIZED	SCHEMA	ASSERTIONS
NVD CVE API 2.0 2026-08-13	curated	Raw JSON	Normalized JSON	Constraints	36
CISA ICS Advisories 2026-08-13	curated	Raw JSON	Normalized JSON	Constraints	51
CISA KEV 2026-08-13	schema-ready	Raw JSON	Normalized JSON	Constraints	12
MITRE ATT&CK 19.1 2026-05-12	curated	Raw JSON	Normalized JSON	Constraints	68
FIRST EPSS v4 not bundled	connector	Raw JSON	Normalized JSON	Constraints	0
MITRE D3FEND 1.4.0 2026-03-31	curated	Raw JSON	Normalized JSON	Constraints	43
RTS-GMLC public benchmark	abstracted	Raw CSV	Normalized JSON	Constraints	38

PIPELINE

Reproducible transformation stages

Acquire

Preserve source bytes and metadata

→

Normalize

Map identifiers to umbrella IRIs

→

Validate

Apply JSON Schema / SHACL

→

Reconcile

Resolve product and asset identity

→

Derive

Generate temporal assertions

→

Publish

Named graph + provenance

Energy Grid-KGs

WG

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INTEGRATED RESEARCH MODEL

V3.0 - WIRG

Named graphs

8

modular ontologies

Integrated entities

252

research snapshot

Schema downloads

8

JSON-LD + SHACL

IT/OT boundaries

4

enterprise to process

Electrical dependencies

17

control and protection

GRAPH FEDERATION

Named graph registry

Deployment & Identity KG

synthetic · 29 entities · 46 relations

ATT&CK & Campaign KG

public · 41 entities · 68 relations

OT Process & Control KG

synthetic · 20 entities · 29 relations

Detection & Mitigation KG

public+scenario · 28 entities · 43 relations

Product & Vulnerability KG

public · 52 entities · 91 relations

Network Reachability KG

synthetic · 24 entities · 31 relations

Grid & Protection KG

benchmark · 25 entities · 38 relations

PROV-O Temporal KG

derived · 33 entities · 74 relations

ATTACK NAMED GRAPH

ATT&CK & Campaign KG

Download schema

Observed adversary behaviour

Best Practices & Limitations

Follow the Science

Numerical Stability: Mandatory manual oversight to catch floating-point degradation or mathematical compounding errors introduced by LLMs.

Explicit Prompting: Forcing clear computational limits, accurate matrix dimension definitions, and mandatory execution time restrictions.

Human-in-the-Loop Validation: Strict code verification and isolated sandbox execution before embedding untrusted algorithms into physical experiments

Cognitive Offloading

Productive vs. Substitutive Offloading: Saving time on minor script syntax tasks versus completely outsourcing core architectural thinking.

The Cognitive Debt Cycle: Over-dependence on AI code generation tools degrades comprehensive long-term architectural understanding.

The Automation Paradox: Transitioning humans purely to code critics strips away the muscle memory needed to spot deep operational errors.

Deliberate Cognitive Friction: Constructing architectural forcing functions like the Hint-First system to guide engineers instead of blindly printing code.